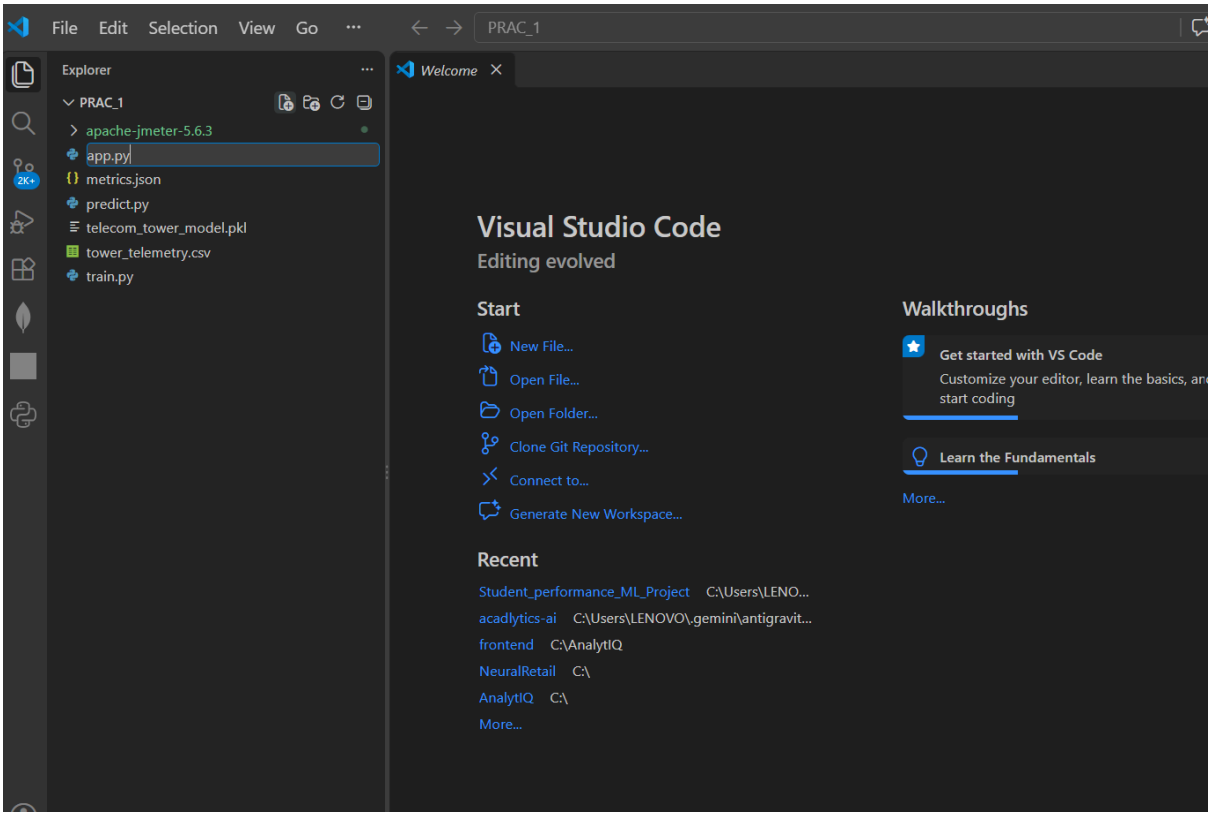
PRACTICAL JMETER - To perform load testing on a Machine Learning API using Apache JMeter and analyze its performance under multiple user requests

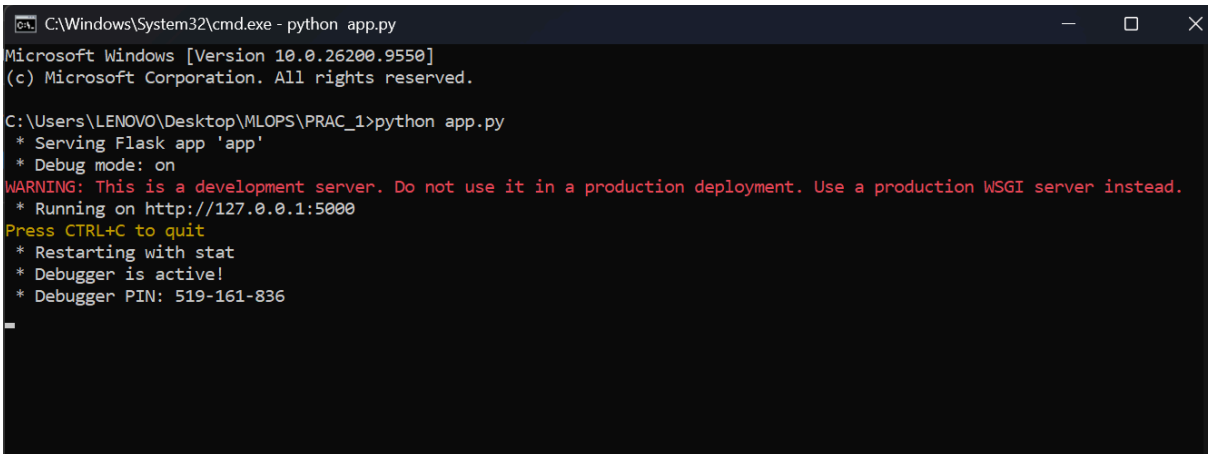
1. Go to cmd type pip install flask
Also check if java installed in cmd java - - version
2. Go to vs code open your Mlops folder and add app.py



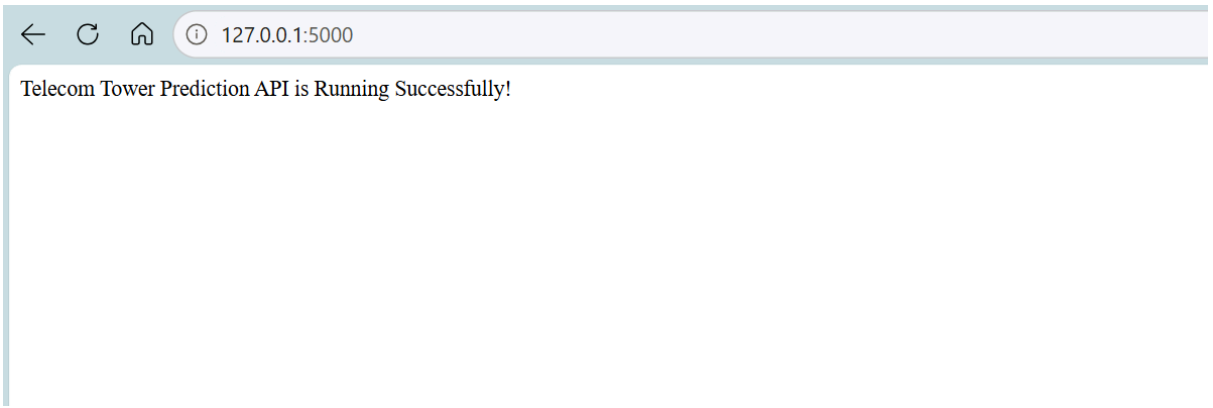
3.Paste this code in app.py

```
from flask import Flask, request, jsonify
import pandas as pd
import joblib
app = Flask(__name__)
model = joblib.load("telecom_tower_model.pkl")
@app.route("/")
def home():
    return "Telecom Tower Prediction API is Running Successfully!"
@app.route("/predict", methods=["POST"])
def predict():
    data = request.get_json()
    df = pd.DataFrame([data])
    prediction = model.predict(df)
    return jsonify({
        "prediction": int(prediction[0])
    })
app.run(debug=True)
```

4 Go to cmd type python app.py
Make sure you are in your file path

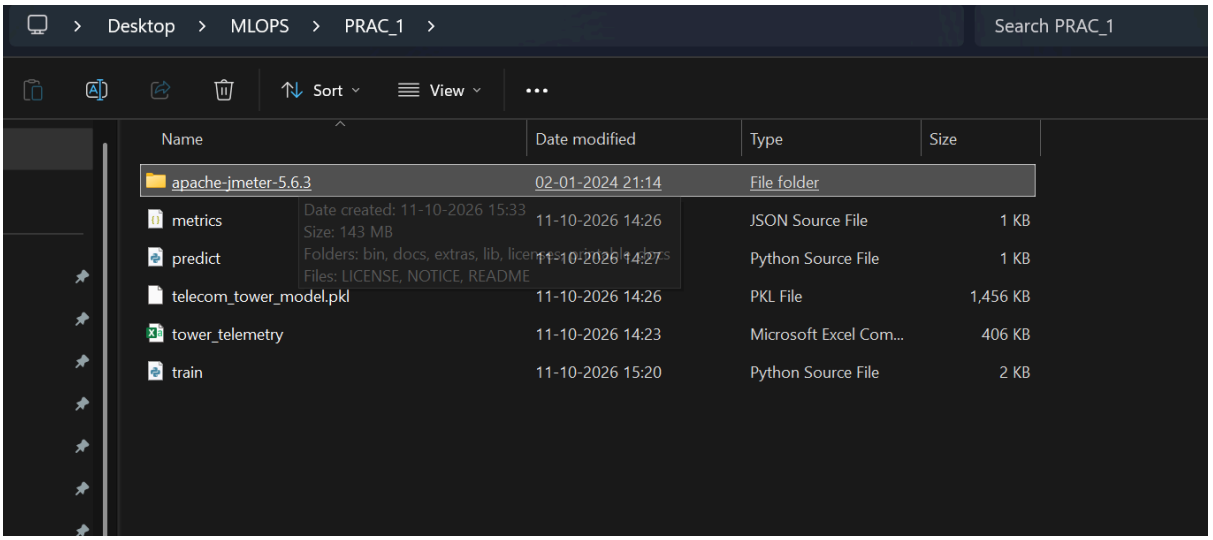


5.open the browser from your cmd output
(copy the http:// and paste it in crome or any browser)

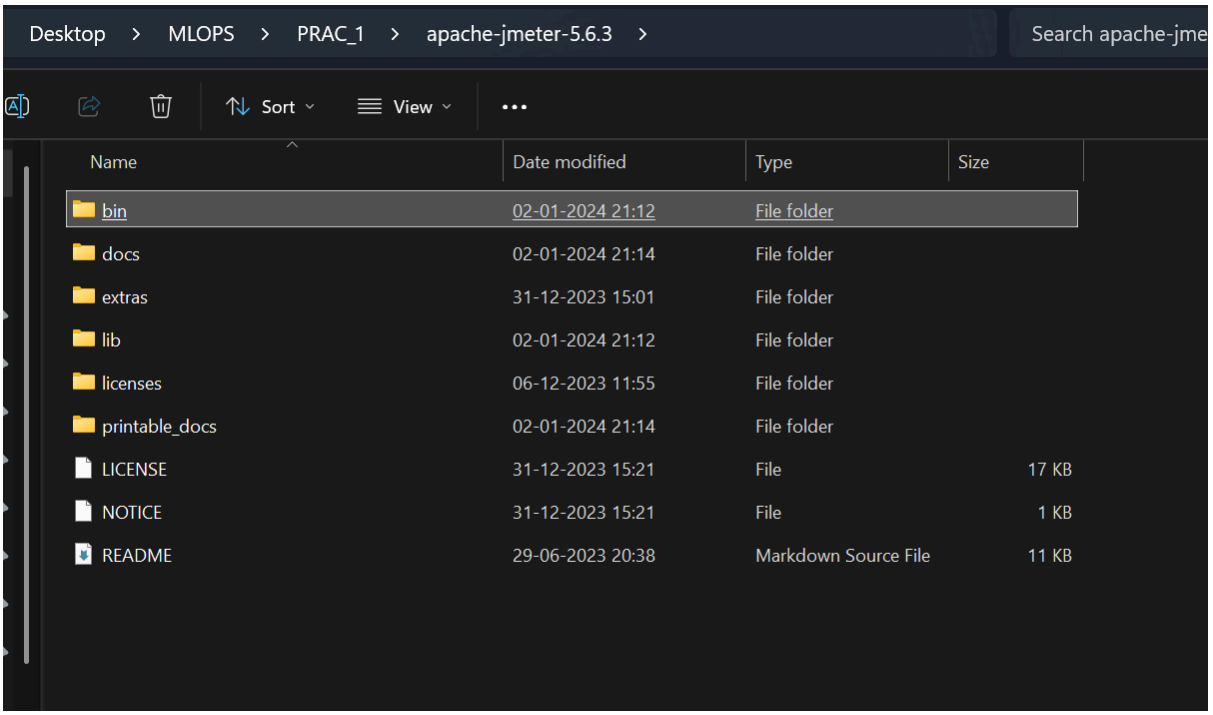


You should see this as output

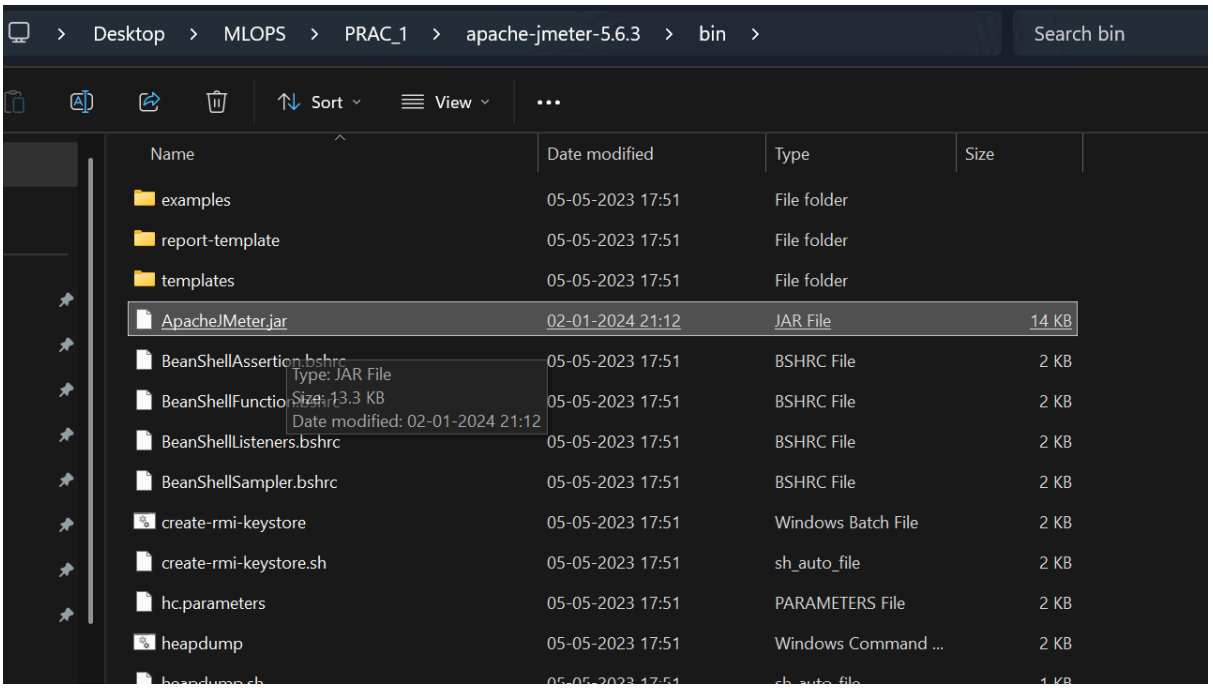
6.install apache jemeter
7. Extract all the zip apache file in your Mlops prac folder only



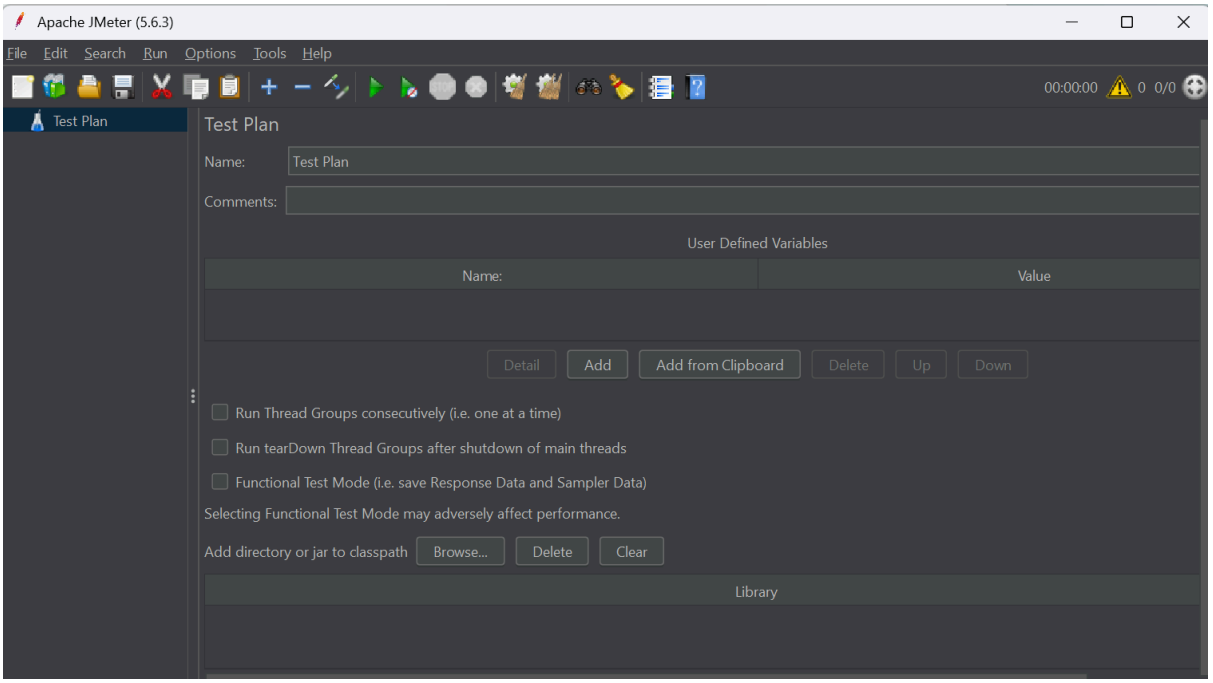
8.Then go in apache jemeter folder select bin



9. In bin then open apache jmeter (Make sure java is installed by cmd java - - version)



10 open apache jmeter.jar

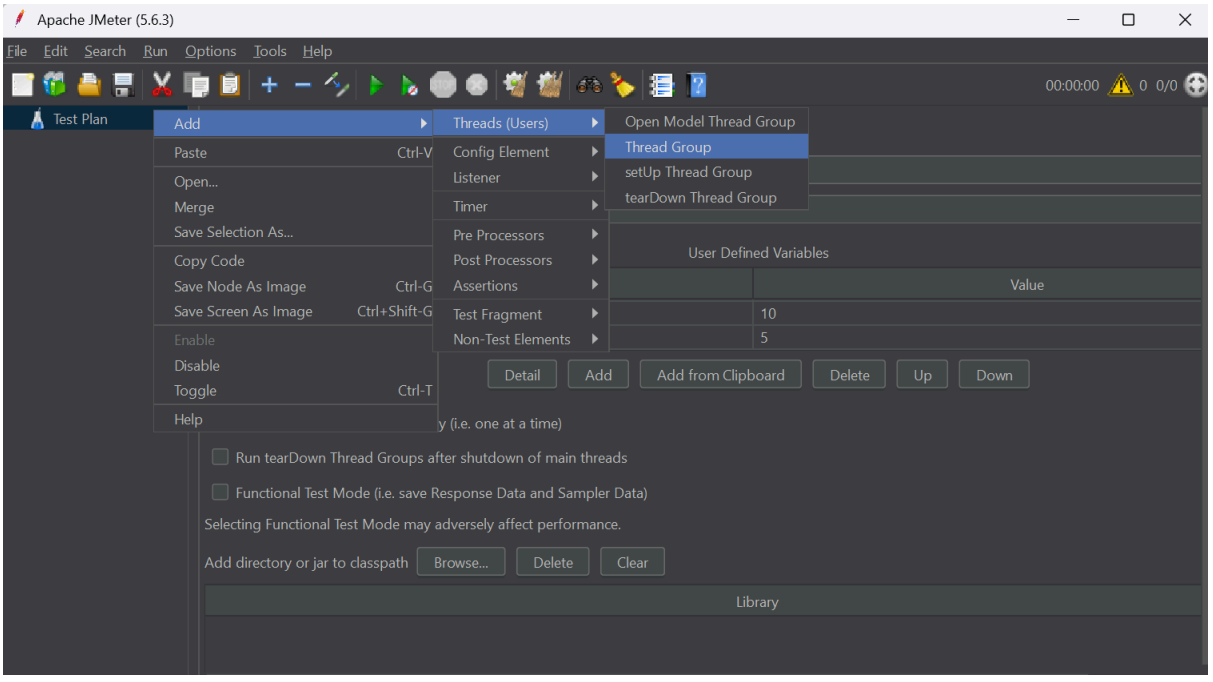


In test plan select add and add under names and value

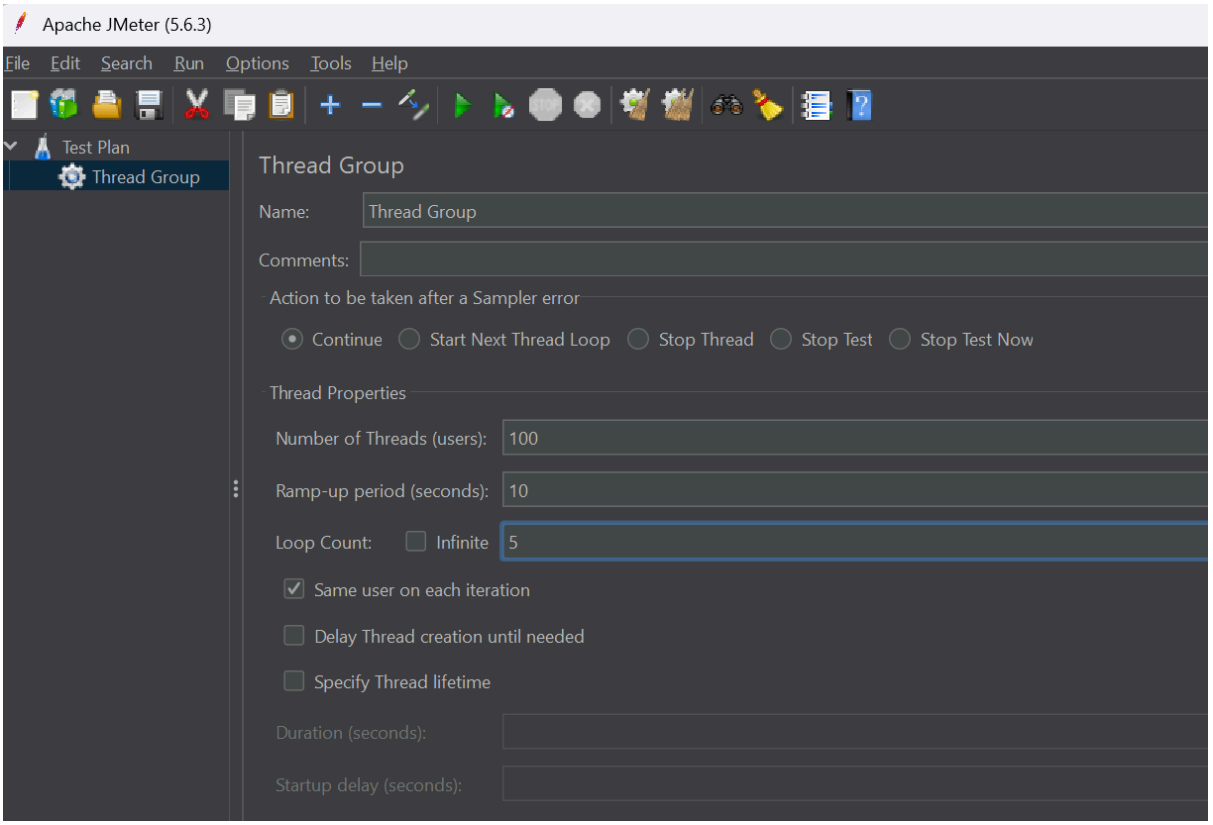
Names	value
Number of Users	100
Ramp-Up Period	10
Seconds Loop Count	5

(also remember to change these numbers i have taken sirs numbers)

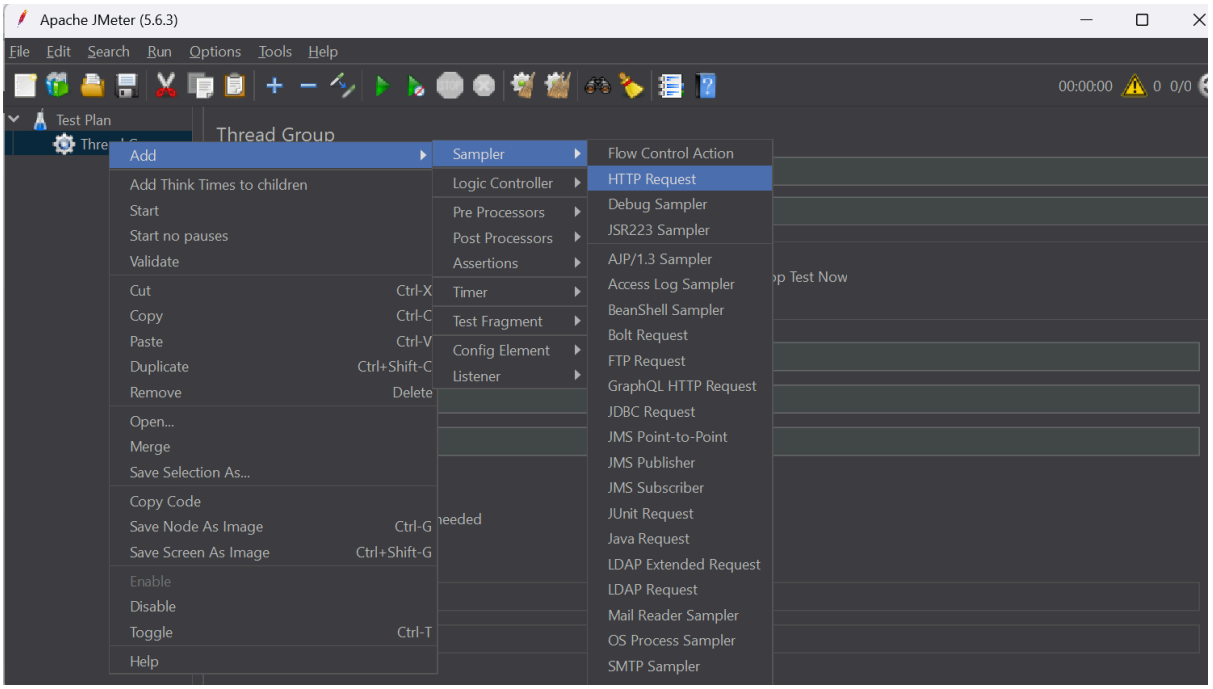
11.add threads group (right click on test plan)



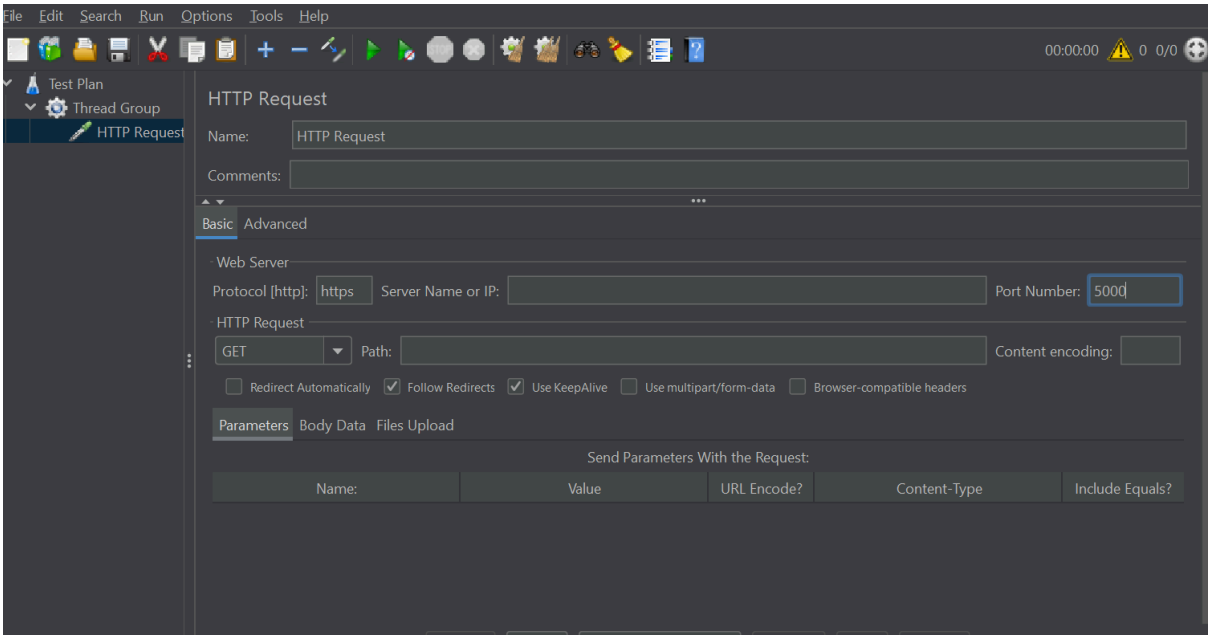
12.then fill up the same values that you had filled in test plan



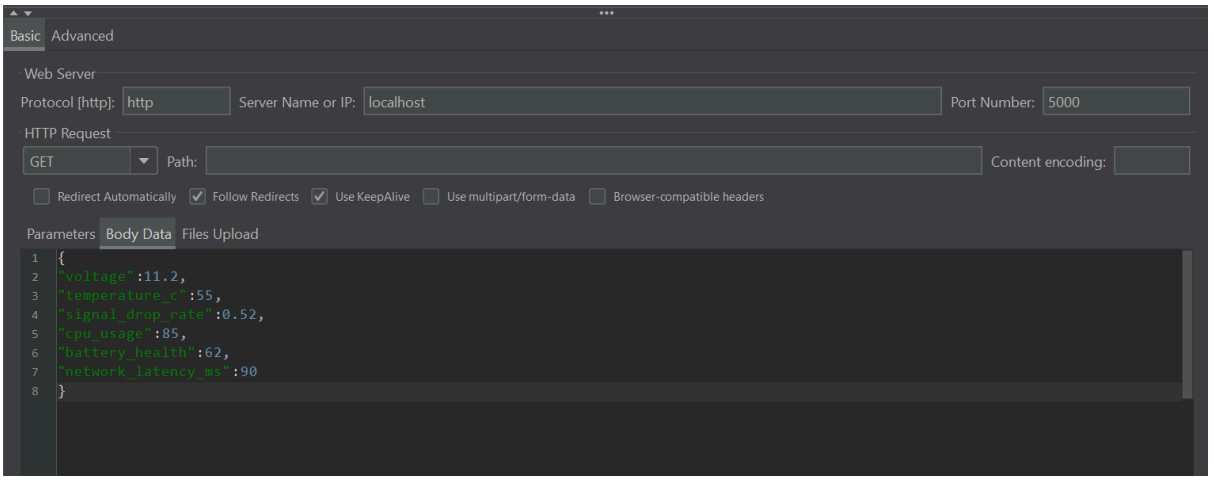
13.next right click on threads group add https request



14. Add - **protocol [http]: https** and **port number : 5000** and **server or IP : localhost** (But incase you get error ahead you can change settings where **HTTPS Request - POST , Path : /predict ,** and do check json code**)**

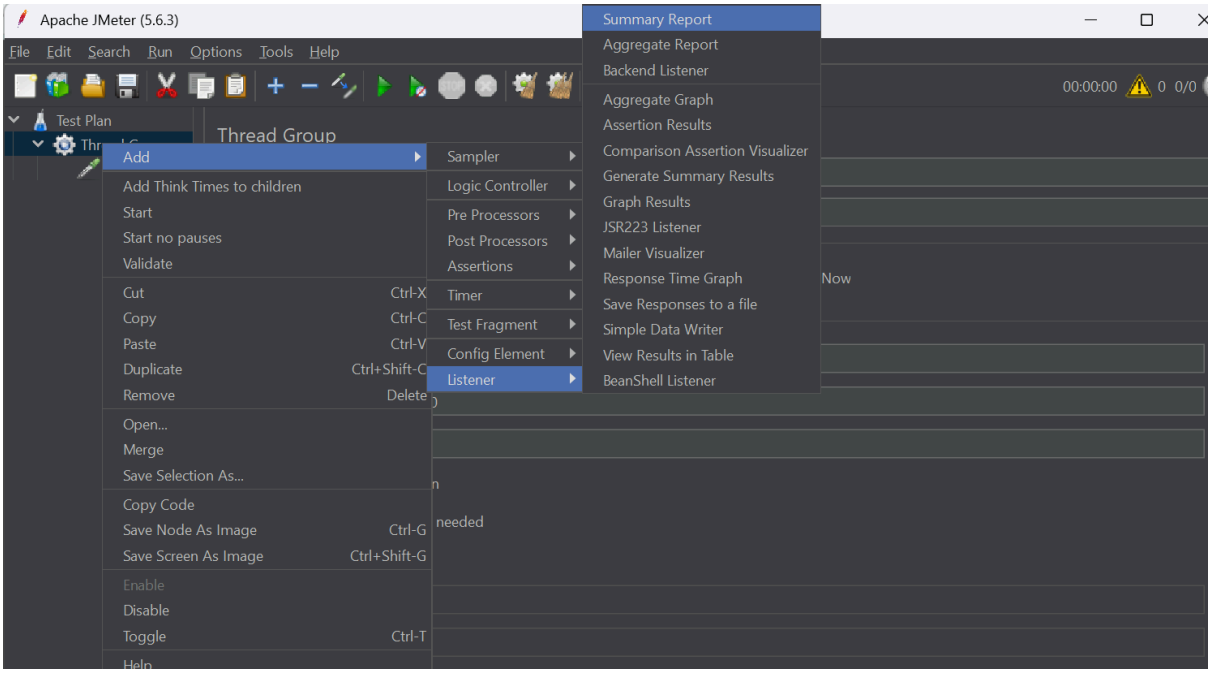


15.next in in https only in body data add this json code

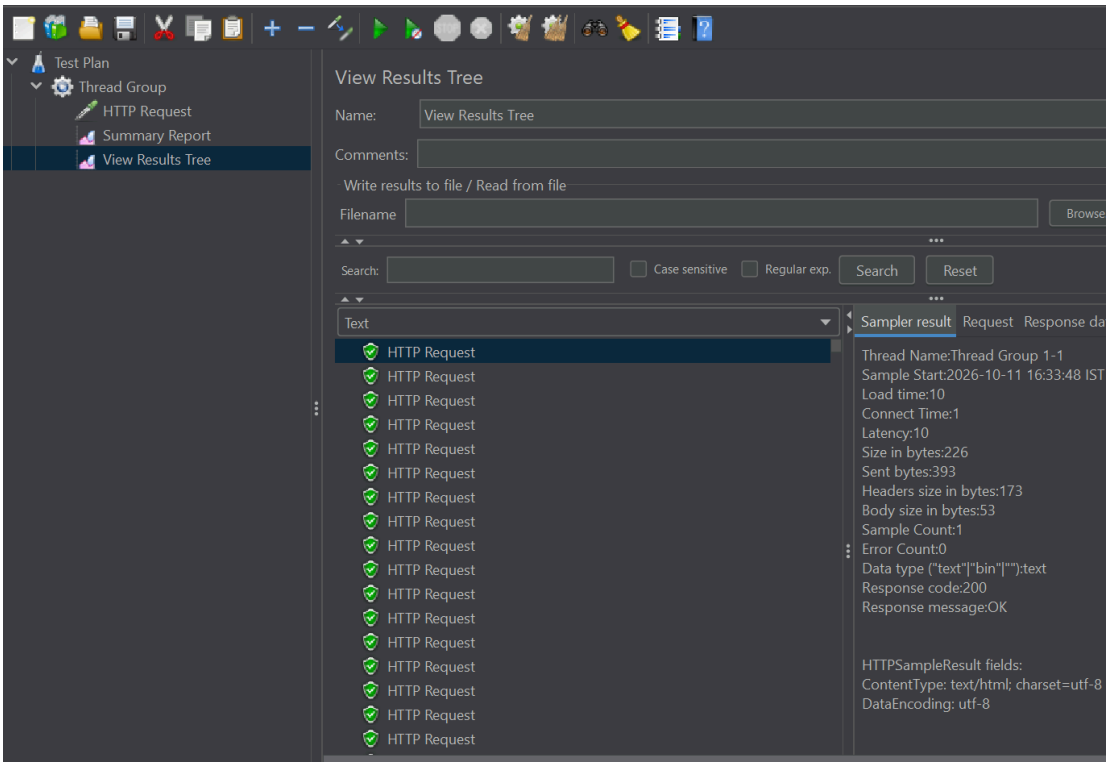


{
"voltage":11.2,
"temperature_c":55,
"signal_drop_rate":0.52,
"cpu_usage":85,
"battery_health":62,
"network_latency_ms":90
}

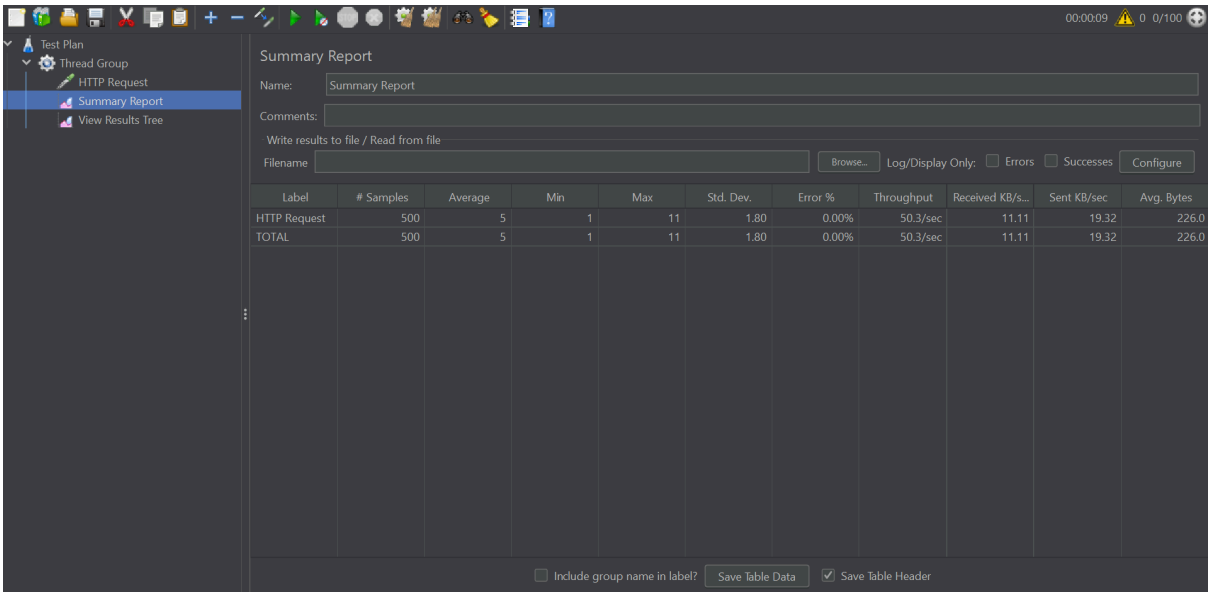
16. Then right click thread group - listener -summary report

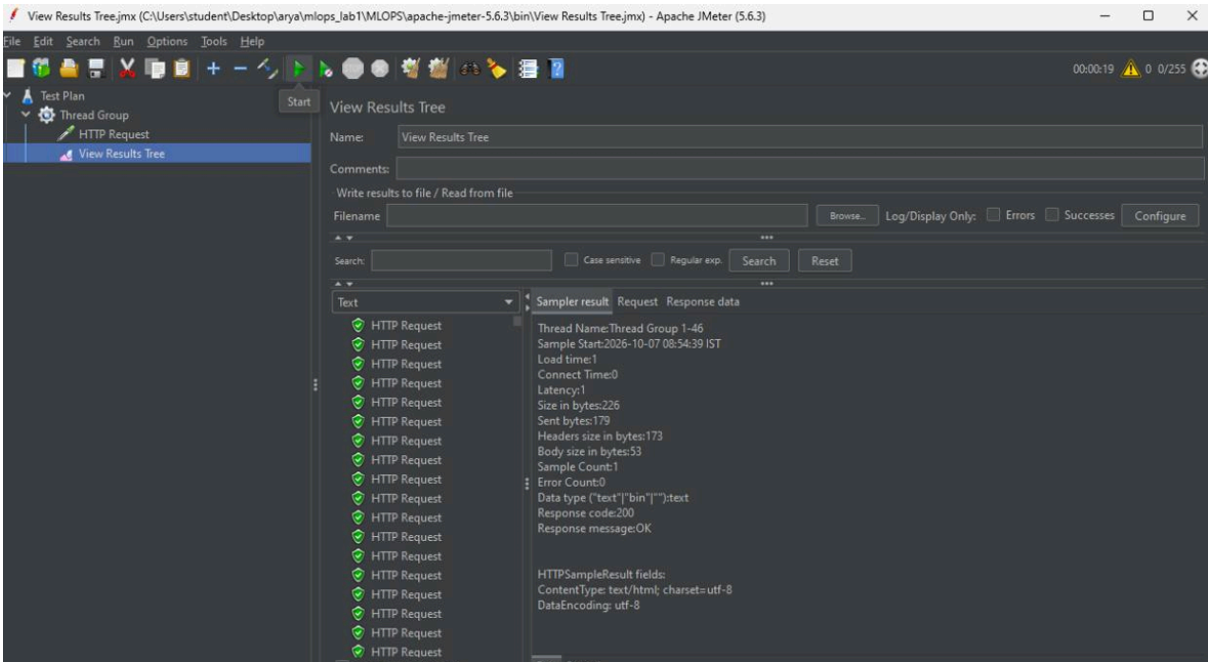


17 similarly add thread group - listener - view result tree
(this should show green if red recheck code and https request)



View output


00:00:09 0 0/100
Test Plan
Thread Group
HTTP Request
Summary Report
View Results Tree
Summary Report
Name: Summary Report
Comments:
Write results to file / Read from file
Filename
Browse...
Log/Display Only: Errors Successes
Configure
Label # Samples Average Min Max Std. Dev. Error % Throughput Received KB/s... Sent KB/sec Avg. Bytes
HTTP Request 500 5 1 11 1.80 0.00% 50.3/sec 11.11 19.32 226.0
TOTAL 500 5 1 11 1.80 0.00% 50.3/sec 11.11 19.32 226.0
Include group name in label?
Save Table Data
Save Table Header


View Results Tree.jmx (C:\Users\student\Desktop\arya\mlops_lab1\MLOPS\apache-jmeter-5.6.3\bin\View Results Tree.jmx) - Apache JMeter (5.6.3)
File Edit Search Run Options Tools Help
00:00:19 0 0/255
Start
Test Plan
Thread Group
HTTP Request
View Results Tree
View Results Tree
Name: View Results Tree
Comments:
Write results to file / Read from file
Filename
Browse...
Log/Display Only: Errors Successes
Configure
Search:
Case sensitive
Regular exp.
Search
Reset
Text
Sampler result Request Response data
HTTP Request
Thread Name:Thread Group 1-46
Sample Start:2026-10-07 08:54:39 IST
Load time:1
Connect Time:0
Latency:1
Size in bytes:226
Sent bytes:179
Headers size in bytes:173
Body size in bytes:53
Sample Count:1
Error Count:0
Data type ("text"|"bin"|""):text
Response code:200
Response message:OK
HTTPSampleResult fields:
ContentType: text/html; charset=utf-8
DataEncoding: utf-8